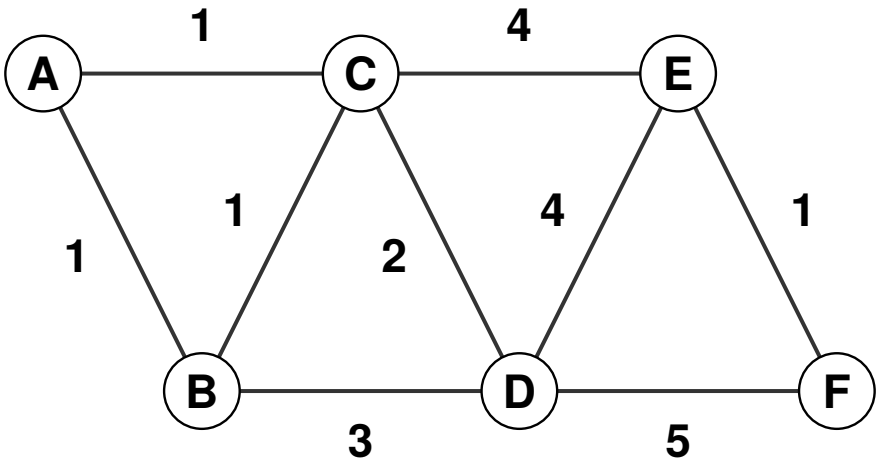
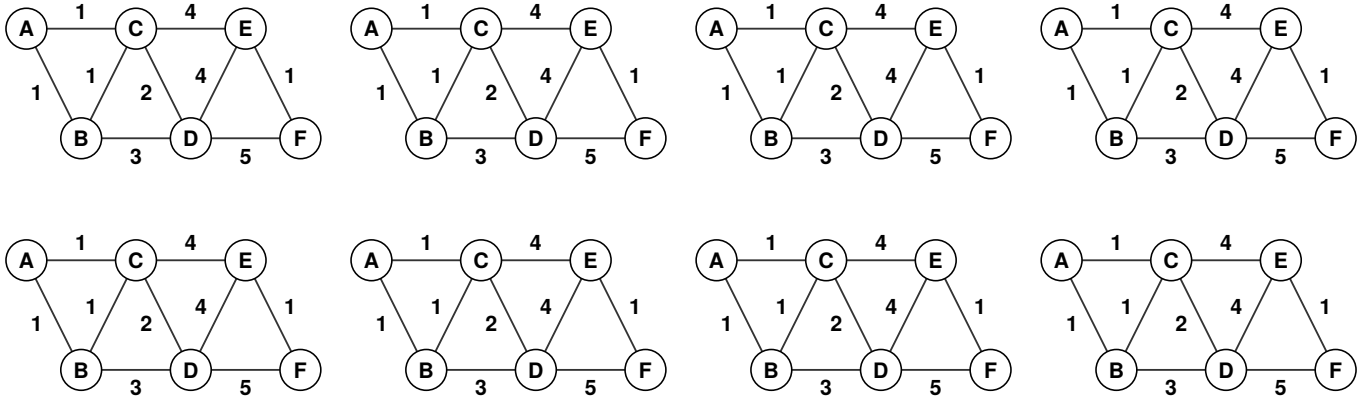
Buy a link by laying a marker along it and putting its price in counters in the tray. A network is a set of bought links that lets the pawn walk on markers between every two places.

Problem 1: Buy the cheapest network you can on this map. How many different networks cost exactly that much? Draw them.





Problem 2: A route is a way for the pawn to walk from one place to another along links, such as ABD. Find a route from A to D whose biggest price is as small as possible. Do the same from B to D, from D to F and from A to F.

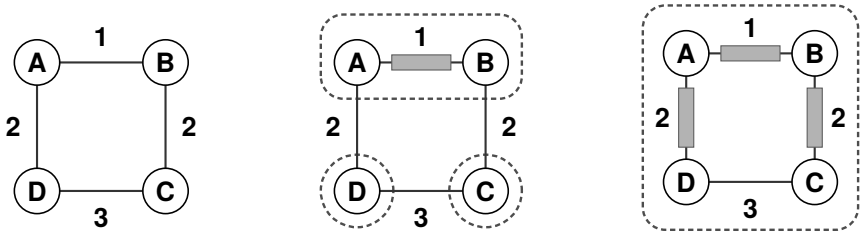
A to D: route _____ biggest price _____

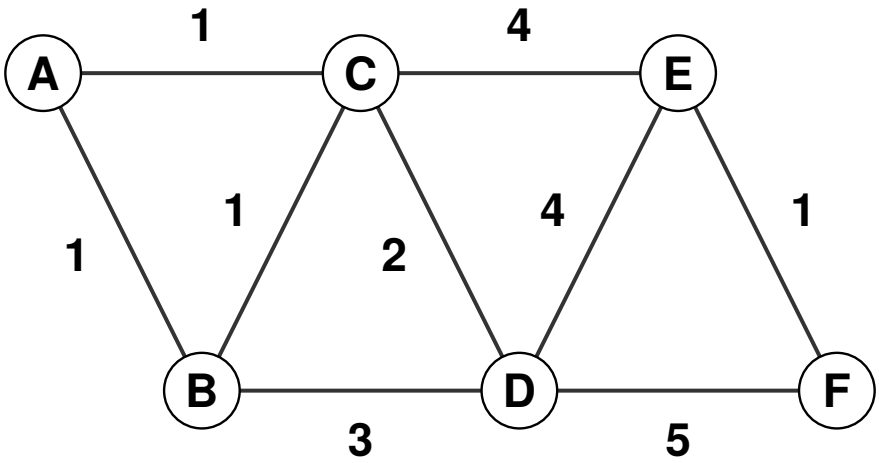
B to D: route _____ biggest price _____

D to F: route _____ biggest price _____

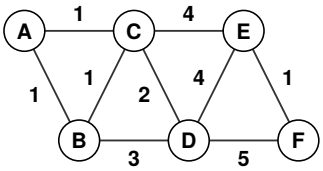
A to F: route _____ biggest price _____

Problem 3: On the big map, lay markers on the links priced 1, and draw a ring around each group of places that the pawn can walk between on markers, as in the small pictures. Then add the links priced 2, then 3, and so on, ringing the groups each time, until there is one group. At which price do A and D first share a ring? B and D? D and F? A and F?

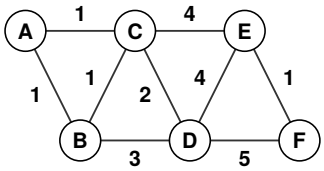




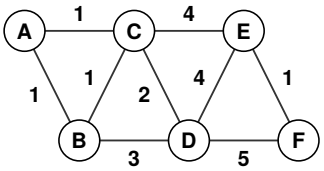
priced 1



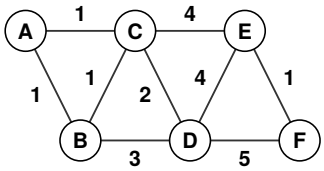
priced up to 2



priced up to 3

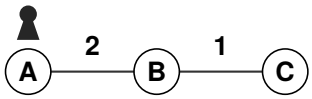


priced up to 4



A and D: _____ B and D: _____ D and F: _____ A and F: _____

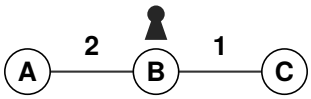
Problem 4: On a trip, the pawn walks from one place to another along any links, and you pay each link's price every time the pawn walks along it. Write the trip by its letters, and pay when the trip is over.



Trip: **A**



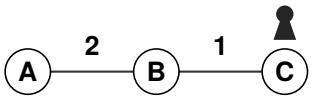
Start



Trip: **AB**



Walk to B



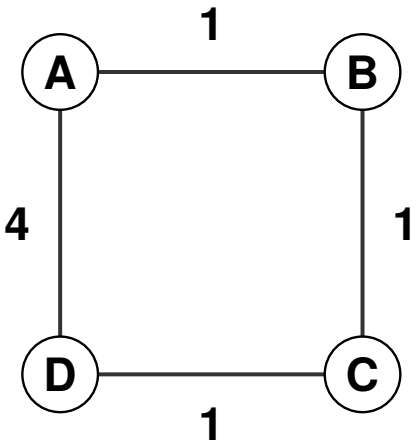
Trip: **ABC**



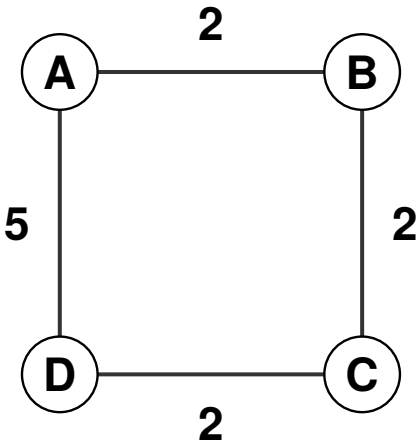
Walk to C, then pay

On each map, find the cheapest network and the cheapest trip from A to D.

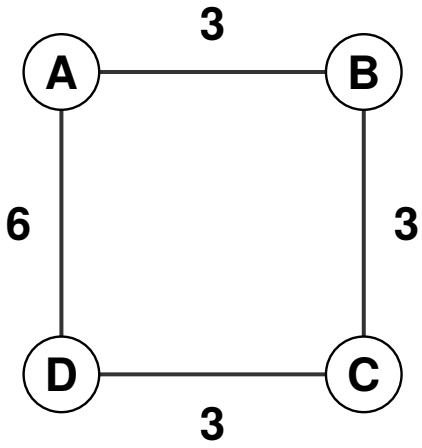
Original prices



Every price 1 more



Every price 2 more



Network _____
costs _____

Trip _____
costs _____

Network _____
costs _____

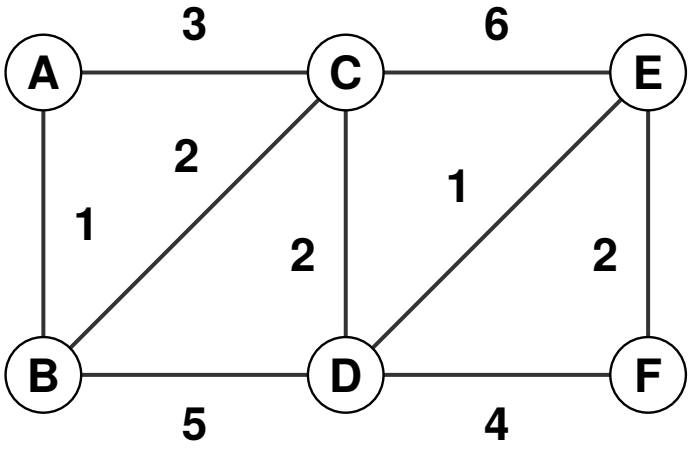
Trip _____
costs _____

Network _____
costs _____

Trip _____
costs _____

Problem 5: On each map, find the cheapest network and every cheapest trip from A to F.

Original prices



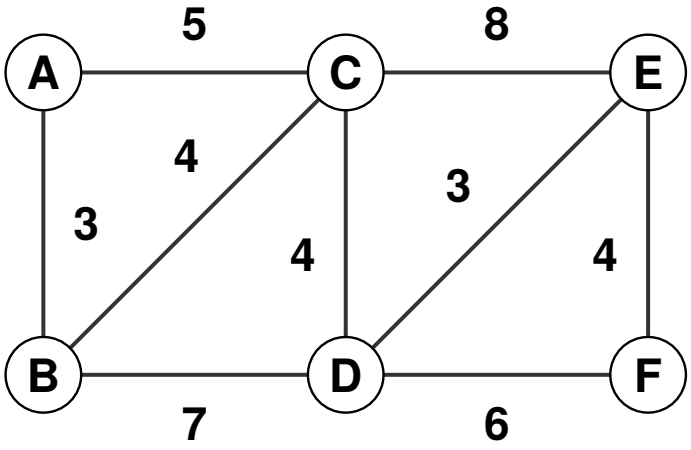
Network _____

costs _____

Trips _____

each cost _____

Every price 2 more



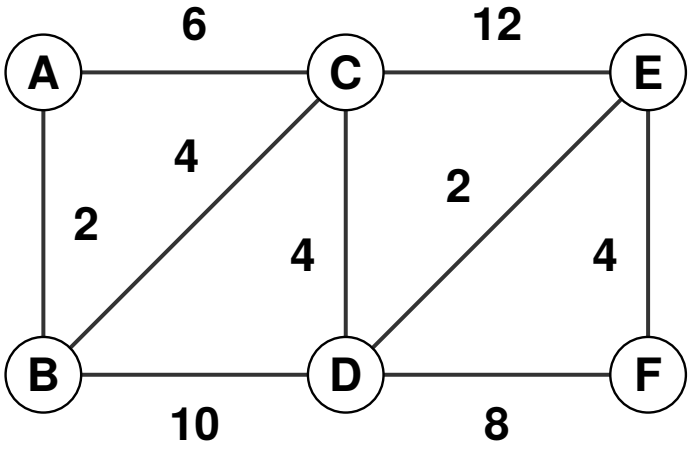
Network _____

costs _____

Trips _____

each cost _____

Every price doubled



Network _____

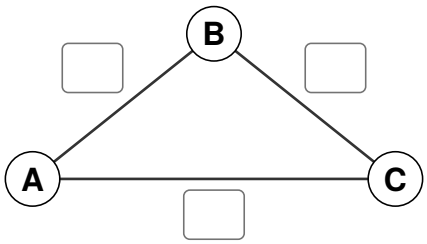
costs _____

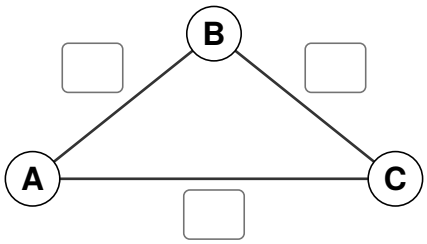
Trips _____

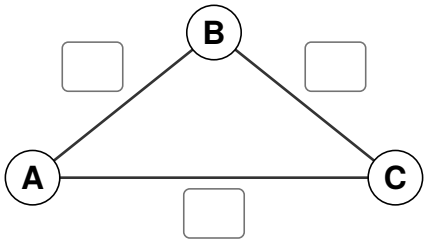
each cost _____

Problem 6: Choose prices for this map so that the cheapest trip from A to C goes through B, but when every price is 2 more, the cheapest trip goes straight from A to C. What must be true about prices that work?

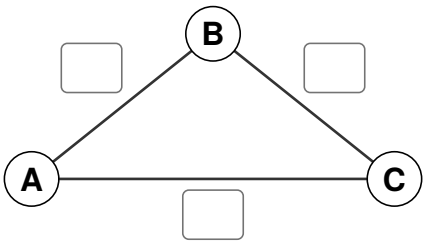
Your prices

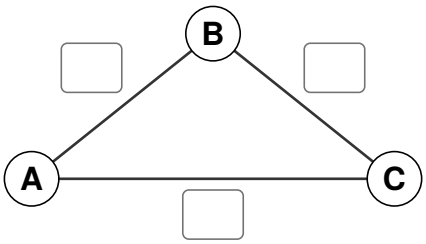


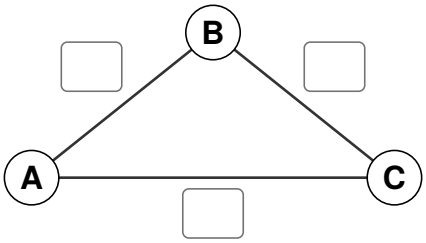




Every price 2 more

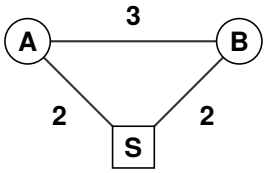




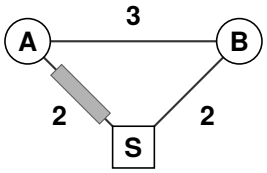


Problem 7: Can adding the same number to every price of a map ever change which networks are cheapest? Can doubling every price ever change the cheapest trips? Explain.

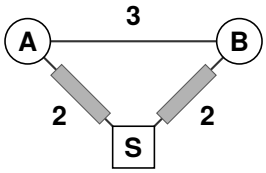
Problem 8: On these maps, a network must join the circled places, and it may go through the place in a square or leave it out. On each map, find the cheapest network, once without the square and once using it.



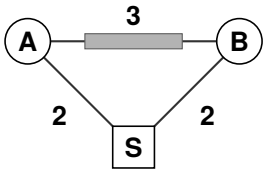
Start



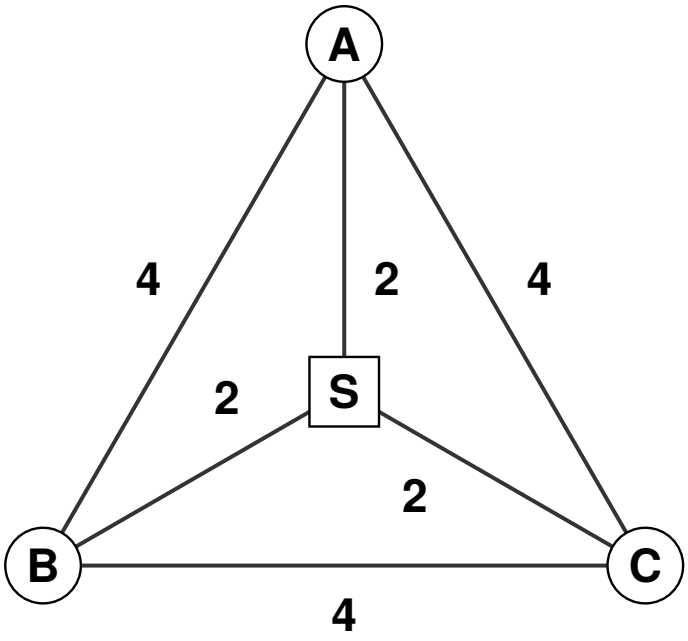
Buy AS



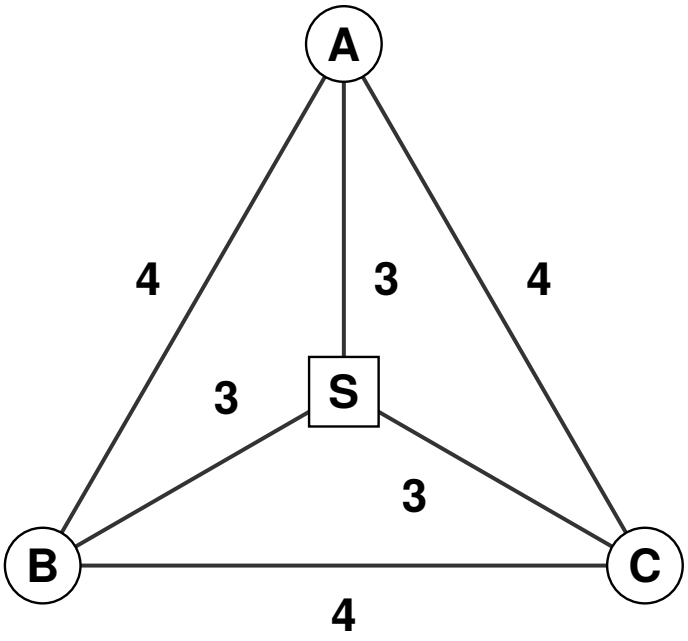
Buy SB



Or buy only AB



without ☐ using ☐



without ☐ using ☐